

Why GCells

After legalization you have legal cell sites, but routers still think in coarser tiles called GCells

The idea

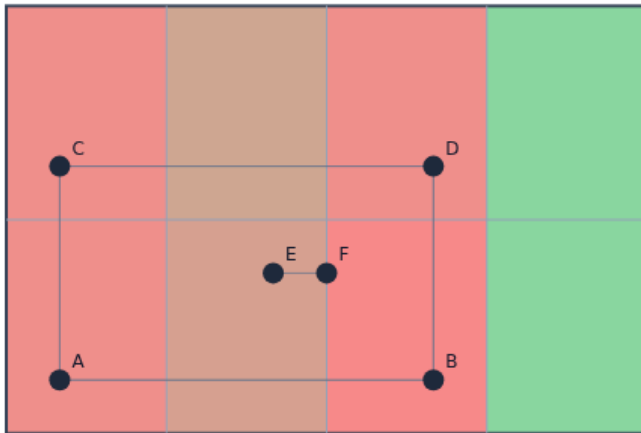
- Index with lower-left origin
- Column i is floor of x over cell width, clamped to zero through three
- Row j is floor of y over cell height, clamped to zero or one
- Cell A at one comma one lands in GCell zero comma zero
- Cell D at eight comma five lands in two comma one
- Memorize that mapping, every estimator reuses it

Chip and GCells

GCELL GRID

Chip and GCells

Step 1 / 5 · chip · module01-01-gcell-grid



Explanation

Twelve by eight chip with a four-by-two GCell overlay.
Each tile is three by four.

- Lower-left origin
- $i = \text{floor}(x/3)$
- $j = \text{floor}(y/4)$

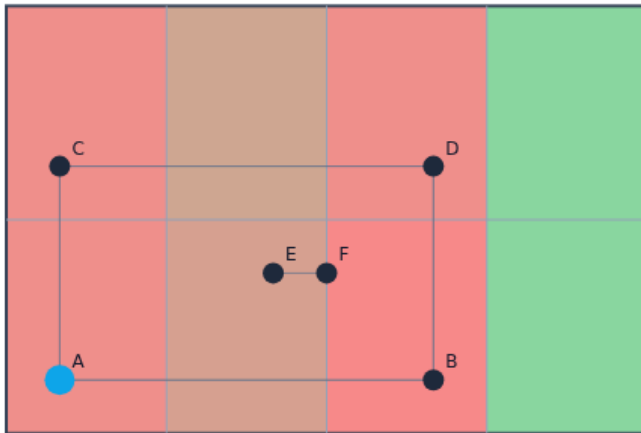
Grid: 4×2
cellW=3 cellH=4

A lands in (0,0)

GCELL GRID

A lands in (0,0)

Step 2 / 5 · a00 · module01-01-gcell-grid



Explanation

Cell A at (1,1) indexes to GCell column 0, row 0.

- Clamp to last tile on edges

A → (0,0)

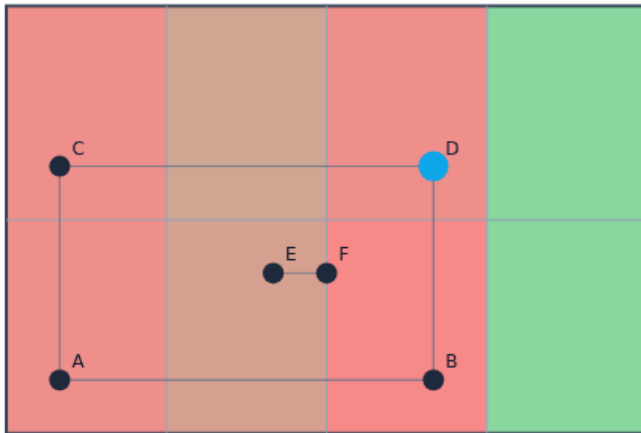
A lands in (0,0)

D lands in (2,1)

GCELL GRID

D lands in (2,1)

Step 3 / 5 · d21 · module01-01-gcell-grid



Explanation

D at (8,5) → $i=2, j=1$ (top row).

- Top row is $j=1$

D → (2,1)

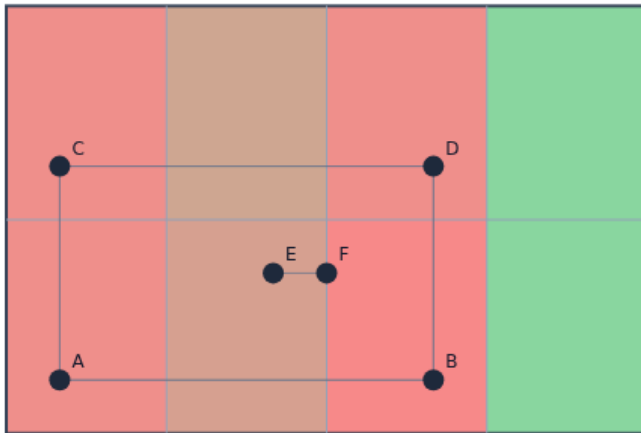
D lands in (2,1)

Paint all centers

GCELL GRID

Paint all centers

Step 4 / 5 · paint · module01-01-gcell-grid



Explanation

Every cell maps to exactly one GCell; the grid is the router's coarse map.

- Used by every estimator

8 tiles total

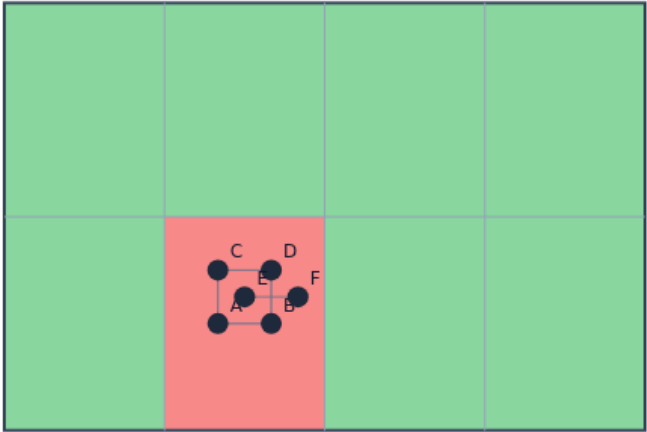
Paint all centers

Clustered seed

GCELL GRID

Clustered seed

Step 5 / 5 · seed · module01-01-gcell-grid



Explanation

Congested seed piles cells into center tiles—later labs heat those GCells.

- Same grid, hotter map

Center columns

Clustered seed

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / GCell grid model

INTERACTIVE TOOL

GCell grid model

Map cell centers onto a 4×2 GCell grid over a 12×8 chip.

Your job: click to place cells. Challenges check GCell indices of *your* positions (A → (0,0) on spread starter).

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] A in (0,0)

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F←→↑↓Reset to starterReveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (0,0)
HPWL: 34.0
overflow total: 22.00 · max: 4.67 · count: 6
capacity: 2
view: your placement

Browser lab starter

learn_congestion — 01 gcell grid

Implement track

- Parse ``tiny_cong.json``
- Print `nx, ny, cellW, cellH`
- Write ``cell_gcell(x, y)`` returning `(i, j)`
- Assert $A \rightarrow (0,0)$ and $D \rightarrow (2,1)$ on the spread placement
- No demand yet, this module is pure geometry

Pitfalls

- Using upper-left image coordinates instead of chip lower-left
- Forgetting to clamp points on the right or top edge into the last tile
- Mixing site columns from legalization with GCell columns, different grids

Your turn

- Finish the checklist
- Be able to sketch the eight GCells from memory
- Next: capacity versus demand, the supply side of the map